

EV CHARGING STATION AVAILABILITY TRACKING DATASET

Dataset Structure and Important Features

Feature Category	Features (Column Name)	Feature Examples	Description
 1. Temporal Features	- timestamp - hour_of_day - day_of_week - is_weekend	2024-07-01 14:00:00 14 Monday 1 (Yes)	Capture the time-related patterns of charging station usage including time of day, day of week, and weekend indicators.
 2. Operational Features	- ports_available - ports_occupied - total_ports - utilization_rate (%)	3 5 8 62.5	Describe the real-time operational status of the charging station including availability, occupancy, and utilization level.
 3. Environmental Features	- temperature_f - humidity (%) - gas_price_per_gallon	86.0 55.0 3.45	Environmental and external conditions that can influence EV charging behavior and station availability.
 4. Waiting Features	- estimated_wait_time_mins - queue_length	12 2	Capture waiting and queue-related information that affects user experience and station demand.
 Target Variable	station_status (Target)	1 (Available) / 0 (Not Available)	Binary target indicating the availability status of the charging station.
Dataset Overview	 Total Instances 10,000	 Total Features 11 (10 predictors + 1 target)	 Target Distribution Available: 6,192 (61.9%) Not Available: 3,808 (38.1%)  Data Source Kaggle (EV Charging Station Availability Tracking)

Figure: Dataset structure showing important feature categories and the target variable used for EV charging station availability prediction.